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Intellectuals, Ideas, and Support for the French Revolution

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SUMMARY

This paper aims to test the longstanding hypothesis of whether the Enlightenment contributed to the French Revolution. To do this, the paper leverages variation in the number of books (novels and philosophy books) and intellectuals as well as several measures of support for the French Revolution: number of political societies, presence of conflicts in 1789, the share of refractory clergy, and the demands in the Cahiers de Doleances. By doing this, the paper makes a significant contribution by providing the first comprehensive tests on the intellectual origins of the French Revolution. Results show that exposure to Enlightenment books and the presence of intellectuals are strong predictors of revolutionary activity. These findings highlight the power of ideas in shaping mass mobilization and institutional upheaval, offering new insights into the ideological foundations of modern political revolutions.

Keywords: Enlightenment, French Revolution, Political ideas, Intellectuals, Book circulation, Institutional change

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1. INTRODUCTION

The influence of ideas on social, political, and economic change has been a central theme in the social sciences, with a long tradition of scholars exploring how intellectual movements shape societal transformations. Weber (1905) reflected on how religious ideas contributed to the rise of capitalism in Western Europe, highlighting the significance of ideational factors in shaping economic behavior.

In another foundational work, Tocqueville (1856) emphasized the transformative role of intellectuals in the lead-up to the French Revolution. Tocqueville argued that the spread of Enlightenment ideas, particularly those promoting individual rights and challenging the legitimacy of absolute monarchy, was crucial in fostering the political climate that allowed the Revolution to unfold. According to his work, the intellectual debate played a key role in shifting public opinion. Intellectuals disseminated revolutionary ideas, contributing to the disintegration of old beliefs, and provided the intellectual legitimacy for radical change.

The role of ideas has also been investigated in the field of economic history. Prominent scholars such as Mokyr (2011) have argued that the Industrial Revolution was as much a product of shifts in knowledge and technological understanding as it was of material factors. Similarly, McCloskey (2010) contends that the rise of capitalism was driven in large part by changes in the ideas and rhetoric surrounding markets, virtue, and entrepreneurship.

The goal of this paper is to empirically test the role of ideas, particularly those stemming from the Enlightenment, in shaping support for the French Revolution. Building on a rich historiographical tradition, this study seeks to deepen our understanding of how intellectual movements influenced political and social change in revolutionary France. Even though Tocqueville (1856) emphasized the transformative role of intellectuals and Enlightenment ideas in precipitating the French Revolution almost two centuries ago, the actual influence of ideas on the Revolution remains a subject of considerable dispute among historians (Burrows, 2014).

The French Revolution stands as a pivotal moment in history, fundamentally reshaping the political and social order of its time. It championed the universal principles of human rights, dismantled deeply entrenched estate systems, and asserted the equality of all individuals before the law. The Revolution also laid the groundwork for the separation of church and state, breaking centuries-old ties between religious institutions and politi-

cal power, and fostering the ideals of secular governance. These transformative changes not only redefined the relationship between individuals and the state but also set a global precedent, inspiring movements for freedom, justice, and equality across the world. These distinctive features of the French Revolution, combined with the scarcity of quantitative analyses of the phenomenon, make it a compelling case study for examining the role of ideas as a precondition for institutional change: the formation of support for such change.

In this paper, I will test the role of ideas in multiple ways. First, focusing on the direct role of intellectuals as key figures in propagating the diffusion of new ideas. Second, I address the role of books, mainly philosophy and novels (Hunt, 2007; Israel, 2015), in inspiring value change.

To do this, I rely on a broad array of data sources. The presence of intellectuals is identified using biographical information (Laouenan et al., 2022). For books, I use sales data from the *Société Typographique de Neuchâtel* (Burrows, 2020), a major book supplier to the French market, drawing on information about the category of each book and the destination of each purchase. To measure support for the Revolution, I digitised data on the foundation and location of the political societies, proto-political parties that emerged during the revolutionary period. In parallel, I digitise and employ the *Cahiers de Doléances*, the lists of grievances compiled by all electoral territories (*bailliages*) on the eve of the Revolution for the election of the National Assembly. Thanks to their categorization by content by Hyslop (1934), I am able to extract the grievances and identify demands for political and social change across multiple local assemblies, following approaches used in the existing literature (Squicciarini & Voigtländer, 2015; Waldinger, 2024). Finally, I also incorporate data on violent actions carried out in 1789 (Chambru & Maneuvrier-Hervieu, 2024).

Econometrically, I test the role of ideas by exploiting geographical variation in the presence of intellectuals, book circulation, and support for the Revolution. To address endogeneity concerns, I implement several strategies. First, to address potential self-selection in book purchasing and reading, I employ an instrumental variable approach using distance from Neufchâtel—the location of the STN's printing operations—as an instrument for exposure to books. The identification strategy builds on a gravity model of trade, where distance reduces trade intensity, and leverages Neufchâtel's peripheral position in French history, a small Swiss city, to support the exclusion restriction. Second, I conduct falsification tests using the Cahiers de Doléances, examining whether the presence of intellectuals predict demands for economic liberalization, following Squicciarini and Voigtländer (2016). Finally, I changed book categories to ensure that the mechanism is directly related to the content of the books, and not merely to reading books in general.

The results suggest that Enlightenment-related measures are generally significant predictors of support for the Revolution, although their effects are not uniform across all outcomes. In particular, the presence of intellectuals is positively associated with both the incidence of revolutionary conflicts and the formation of political societies. Meanwhile, in the instrumented specification, the diffusion of novels and philosophy books is linked to a higher likelihood of revolutionary conflict, as well as a decrease in counter-revolutionary activity.

This project aims to contribute to the literature in multiple ways. First, it seeks to provide new quantitative insights into the factors driving support for the French Revolution. Despite its recognition as a pivotal turning point in European history (Acemoglu & Robinson, 2013; Roland, 2004), quantitative knowledge of this event still remains limited (Markoff, 1986; Ottinger & Rosenberger, 2023; Squicciarini & Voigtländer, 2016; Waldinger, 2024). Moreover, this study highlights the role of high-culture individuals as political actors (Kurzman, 2008), and evaluates the empirical basis of claims regarding the impact of intellectuals on support for institutional change. Finally, by evaluating these arguments empirically, the project sheds light on the process of modern institutional formation, considering the role of intellectuals, whose importance for economic growth has already been emphasized by multiple studies (Mokyr, 2011; Squicciarini & Voigtländer, 2015).

2. LITERATURE REVIEW

2.1. Ideas in social Science

Ideas have long stood at the center of inquiry in the social sciences. From the work of the classics—most notably Weber (1905), whose analysis traced the roots of modern capitalism to the ethical principles of Protestantism—scholars have repeatedly returned to the role of ideas in shaping social, political, and economic life. Ideas are not mere reflections of material conditions; rather, they serve to define what actors consider legitimate, desirable, and possible. Across disciplines and theoretical traditions, researchers have developed diverse frameworks to explain how beliefs, norms, and ideologies influence both individual behavior and institutional change.

This foundational interest in the power of ideas was further developed by Gramsci (1975), whose theory of hegemony profoundly shaped how scholars understand the relationship between ideas and power. For Gramsci, domination was not sustained solely through coercion but through consent, achieved by embedding ruling-class interests within a broader, seemingly "common sense" worldview. Hegemony, in this sense, involves the diffusion of ideas and values that come to be seen as natural, legitimate, and universal—even by those who might be materially disadvantaged by them. In his work, ideas are central in legitimating and stabilizing power relationships.

This ideational perspective has gained particular prominence in the field of International Relations, where it constitutes the foundation of the constructivist paradigm. Constructivists argue that the international system is not a fixed or objective structure dictated solely by material capabilities such as military or economic power (Wendt, 1992). Instead, it is a socially constructed realm in which norms, identities, and beliefs fundamentally shape the preferences and behaviors of states and other actors. Seminal contributions in this field include the work of Finnemore and Sikkink (1998), who argue that norm diffusion—particularly through mechanisms like norm entrepreneurship, socialization, and institutionalization—can significantly alter state behavior. Finnemore (1993) analysis of international organizations, such as UNESCO, illustrates how the spread of norms regarding appropriate state conduct has reshaped global expectations in areas like education, military intervention, and human rights. These norms gain traction not only through coercion or material incentives but also through internalization, wherein actors come to accept new ideas as legitimate and morally binding.

Another important domain where the impact of ideas has been extensively studied is the abolition of slavery. Seminal work by Fogel and Engerman (1977) in economic history has challenged materialist explanations that slavery was abolished due to economic inefficiency. His research on the American case demonstrates that the Southern economy, which relied heavily on slavery, was in fact more productive than the Northern economy at the time. This finding undermines the notion that the decline in the economic productivity of slaver led to its abolition (Phillips, 1952). Ultimately hinting that abolition was driven by a moral transformation—a shift in the social and ideological acceptability of slavery. In this light, ideational change was decisive in overturning a deeply entrenched and materially profitable institution. Complementary historical accounts have also underscored the importance of normative and religious motivations. For instance, Drescher (1987) emphasizes the role of humanitarian and evangelical ideals in driving the British abolitionist movement. Similarly, scholars have noted the influence of religious groups such as the Quakers in shaping abolitionist sentiment through moral persuasion and public activism (Jones, 1998).

Another significant strand of the literature highlights the role of ideas as key drivers in the adoption and transformation of economic regimes. This scholarship rejects deterministic models of economic change that rely solely on structural constraints or material interests. Instead, it emphasizes how ideational frameworks—such as policy paradigms, intellectual currents, and normative beliefs about markets and state intervention—guide the choices that governments make. Blyth (2002) represents a pivotal contribution to this literature. Focusing on the political-economic transformations in the United States and Sweden, Blyth argues that crises in institutional frameworks open space for ideational change. During such crises, existing institutional arrangements lose legitimacy, and actors search for new interpretative frameworks. Ideas, in Blyth's view, help to define what constitutes a crisis and what kinds of solutions are seen as credible or desirable. He shows how ideational entrepreneurs reframe interests, build political coalitions, and provide new legitimizing narratives for institutional reform. Contrary to theories that explain institutional change solely through shifts in power or material conditions, Blyth's work highlights the autonomous role of ideas in shaping the path and content of transformative change. Moreover, Kus (2024) argues that the neoliberal turn in U.S. economic policy was enabled by the convergence of academic and policy spheres: neoliberal ideas gained dominance in economics departments and were subsequently translated into governance through the rising influence of economists in policymaking roles. Similarly, Quinn and Toyoda (2007) provide cross-national evidence linking ideological shifts to liberalization trends in trade and finance. These works demonstrate that economic regimes are not simply technocratic responses to material conditions but are deeply shaped by prevailing

beliefs and discursive frameworks.

2.2. The Revolution and Ideas

This project engages with a diverse array of contributions across several fields, including history, economic history, political science, and political sociology. The research question is not new, with work reflecting on this question for over two centuries (Chartier, 1991; Darnton, 2009; Israel, 2015; Mornet, 1934; Taine, 1876; Tocqueville, 1856). In Tocqueville (1856)'s analysis of the French Revolution, the role of the *homme de lettres* (man of letters) emerges as both crucial and complex. Tocqueville recognizes that intellectuals played a central role in shaping the ideological climate that preceded the Revolution. These writers, philosophers, and political theorists, who were often part of the Enlightenment movement, helped spread ideas of individual liberty, rational governance, and critique of the *ancien régime*. Figures such as Voltaire, Rousseau, and Montesquieu provided intellectual frameworks that challenged the traditional authority of the monarchy and the church.

It also draws on the historical work by Mornet (1934) on the intellectual origins of the revolution, where the author traces the origins of the key ideas of the Enlightenment and tries to look at the diffusion of these ideas throughout the Eighteenth century among different classes of the French population. This seminal work has encountered different critiques, notably on the uniqueness of interpretation of such messages that were spread and on the question of to what extent they were actually followed (Chartier, 1991) and whether books were read (Darnton, 2009). At the same time, it is arguable that the process of the Enlightenment and the Revolution is endogenous to other slow-moving societal changes such as the secularization of French society and mounting fatigue with the Old Regime (Chartier, 1991). Moreover, it draws on two recent works by historians Jonathan Israel and Lynn Hunt. Both books emphasize the pivotal role of intellectual activity, though their proposed mechanisms differ significantly.

Hunt (2007)'s influential book, *The Invention of Human Rights*, highlights the transformative role of novels in catalyzing movements advocating for human rights reforms. In the context of the French Revolution, Hunt argues that the proliferation of novels in the eighteenth century helped cultivate empathy by portraying the shared humanity of individuals through their inner lives. To support this claim, she examines the impact of popular novels from the era and provides qualitative evidence of their influence.

While Hunt's work is groundbreaking, it leaves certain questions unanswered. For instance, the absence of a comprehensive assessment of novels' influence raises doubts

about the extent to which they truly shaped the development of human rights. Moreover, her approach makes it challenging to discern whether the critical factor was the act of reading novels, or broader trends in literacy, or whether the rise of novels coincided with other societal transformations. These could include the emergence of individualistic values (Henrich et al., 2010) and a broader secularization process (Cabello, 2024) during the same period, which may have fostered the formation of a new secular ethic that supported human rights.

The second work underpinning this study is the highly debated *Revolutionary Ideas* (Israel, 2015). This book delves into the intellectual foundations of the French Revolution, highlighting the pivotal role of Radical Enlightenment thinkers. Building on an established historiographical tradition (Taine, 1876; Tocqueville, 1856) and on evidence contemporary to the Revolution that underscores the influence of intellectuals on revolutionary movements, the book contends that the ideas of Radical Enlightenment figures significantly shaped the revolution's demands. While their formal participation in political structures, such as the General Assembly, was limited—with only a few individuals elected—their broader intellectual influence was substantial. These thinkers contributed to reshaping France's prevailing beliefs in the years prior to the Revolution, fostering societal transformations that challenged entrenched hierarchies and encouraged a more inclusive political dialogue. The book posits that these shifts ultimately redefined France's collective identity and approach to governance.

Despite its contributions, the book has faced notable criticism. Scholars have questioned its stark distinction between Radical Enlightenment intellectuals and their more conventional Enlightenment counterparts (Coleman, 2016) and pointed to potential biases in its use of historical narrative to verify its claims (Van Kley, 2014). Nevertheless, it presents a compelling narrative of the French Revolution's intellectual origins and offers a framework that, despite its controversies, can be adapted into a testable empirical claim. Testing these claims will ultimately reveal whether they are well grounded in the historical literature.

The importance of alternative models of governance, as highlighted in studies on attitudes toward democracy (Acemoglu et al., 2024), is also evident in the intellectual contributions of Radical Enlightenment thinkers. Despite their limited formal political involvement, these thinkers helped create alternative governance models that catalyzed a shift in France's collective identity and political structures. This shift is similar to findings in research on emigrants to democratic countries, who, after encountering new institutions, are more likely to support democratization in their home countries, further underscoring the transformative power of alternative governance models. Such individuals are statis-

tically more likely to advocate for democratization upon returning home (Acemoglu et al., 2024; Spilimbergo, 2009), suggesting that exposure to new institutional frameworks inspires support for institutional changes aligned with those models.

While much of this literature focuses on immigration, a parallel mechanism has been proposed regarding the role of intellectuals in driving societal change (Bulutgil, 2022). Intellectuals are seen as key figures in developing new frameworks of legitimacy and governance. By disseminating their ideas, they can channel popular grievances into demands for structural changes that align with these new models. Further research has outlined the conditions under which intellectuals most effectively contribute to democratization. Specifically, intellectuals' influence is greatest when they form a collective identity and are recognized as leaders of a democratic movement by other societal groups (Kurzman, 2008). This suggests that, similarly to emigrants, intellectuals can instigate significant societal transformation by shaping the discourse and advocating for new political systems that challenge existing power structures.

Quantitative research on the French Revolution remains relatively limited, though it has grown significantly in recent years. While the importance of support for the Revolution has garnered increasing attention, many questions remain unanswered. Existing studies have explored the proximate causes behind the demand for institutional change (Waldinger, 2024). Following a broader tradition in historical scholarship (Lefebvre, 1983), this study examines the severe droughts of 1788, arguing that these events were perceived as evidence of the inadequacies of French institutions, thereby fuelling calls for reform.

Another noteworthy contribution investigates the role of Rochambeau's combatants, a French military contingent deployed in New England during the American Revolutionary War (Ottinger & Rosenberger, 2023). The paper argues that exposure to the institutions of the emerging American state influenced these soldiers, who later supported the French Revolution in various capacities.

A third significant study (Squicciarini & Voigtländer, 2016) analyzes the impact of knowledge elites on demands for educational reform and the long-term effects of these changes throughout the nineteenth century. According to the authors, these elites, inspired by Enlightenment ideals, advocated for educational reforms reflected in the *Cahiers de Doléances*. Their efforts ultimately modernized the localities where they resided, leaving a lasting legacy.

A final noteworthy contribution (Markoff, 1986) assesses the linkage between literacy and the uprise of violence in the years 1789 and 1790. The findings of this paper reveal that while literacy is not predictive of violence in itself, there is a significant difference

in the type of violence, with areas with higher literacy rates performing violence against symbols of the regimes instead of taking part in the events of the Great Fear.

While this project shares some thematic similarities with Squicciarini and Voigtländer (2016), it differs in several critical aspects. First, this project focuses broadly on support for institutional change, rather than being confined to educational reforms. Second, although both studies consider knowledge elites as a central factor, this project emphasizes intellectuals as creators of transformative ideas, rather than merely as advocates for change. Finally, the metrics for the dependent variable diverge: this project employs diverse measures of support for institutional change, whereas the knowledge elites' study centers on demands for reform within the Cahiers de Doléances.

This paper also engages with the emerging literature on the role of culture in shaping institutional persistence and change (Alesina & Giuliano, 2015; Bisin et al., 2023; Tabellini, 2008). While much of this research emphasizes the processes of cultural transmission and their effects on institutional structures—and vice versa—this paper focuses on a critical precondition for institutional transformation: the support for change. By examining this dynamic and situating it within a pivotal moment in European history (Roland, 2004), the study seeks to illuminate the role of intellectuals in altering cultural norms, thereby fostering institutional change. Referring to this literature, this project also adds to the debate on the historical forms of legitimacy (Bisin et al., 2023; Greif & Rubin, 2024) by focusing on the case of the French Revolution, which represents the switch to a secular form of legitimacy for the country.

3. THEORETICAL FRAMEWORK

3.1. On the causal power of ideas

It is often argued that ideas lack causal agency, serving merely as ideological expressions that follow material changes (Marx, 1867). While this perspective has merit, there are compelling reasons to consider ideas as possessing a degree of causal autonomy.

One important consideration is that ideas shape how actors interpret the world, including their perceptions of what is desirable, legitimate, or possible (Béland, 2010). These interpretations inform both the goals that actors pursue and the strategies they adopt. This influence may be particularly significant in periods of uncertainty (Blyth, 2002), when material interests are ambiguous or contested, and actors rely more heavily on ideational frameworks to define their preferences and courses of action. In such contexts, ideas do not simply reflect reality—they help to construct it.

At the same time, without the concept of political legitimacy, it is difficult to fully account for within a purely materialist perspective. In this framework, material conditions evolve to a point where they are no longer compatible with the existing political structures. While this mismatch may indeed drive change, historical evidence shows that institutions and practices often persist well beyond their original material purpose (Roland, 2004). Explaining such persistence—and the eventual disruption of these structures—requires engaging with the concept of legitimacy. Change, in these contexts, depends not only on material contradictions and power struggles but also on the emergence of new ideas that challenge entrenched norms and render previously unquestioned arrangements subject to open reflection and contestation.

Furthermore, even if one adopts a reductionist perspective and treats ideas as endogenous to technological or material changes, as argued for example by Morris (2015), this framework does not help explain this case, which makes it interesting. The French Revolution happened before Industrialisation and before the diffusion of fossil fuels. Moreover, there remains the question of why such changes emerge in specific contexts and not others.

A further point is that even when material conditions may enable change, understanding and defining the precise nature of that change is only possible through the lens of ideas. For instance, suppose the widespread use of fossil fuels enables a more egalitarian social arrangement, as the author suggests. The mere existence of new energy

sources or technologies does not inherently produce concepts like "democracy" or "human rights"—those are normative interpretations constructed through ideational frameworks. Material transformations may set the stage, but it is ideas that supply the vocabulary, values, and institutional models through which change is understood, justified, and institutionalized. In this sense, ideas do not merely follow material developments—they mediate and shape how those developments are perceived and enacted. Without ideas, we cannot even meaningfully describe why a more egalitarian structure manifests as liberal democracy rather than some other form of governance. In other words, even if this perspective is correct in its form, ideas help understand the precise nature of such change, enriching the general framework.

3.2. Mechanisms and expectations

Having outlined the theoretical foundations of this project, this section adapts them into testable hypotheses. In particular, it examines three complementary mechanisms through which Enlightenment ideas may have spread: novels, philosophy books, and intellectuals.

The first mechanism focuses on the diffusion of novels. Hunt (2007) argues that the rise of the novel fostered empathy by allowing readers to identify with characters from diverse social backgrounds. Building on this insight, this paper postulates that areas where novels were more widely diffused developed a stronger sense of empathy, leading to greater support for institutional transformations grounded in the principles of human rights.

H1: Areas where novels are more diffused will support more institutional change.

The second mechanism concerns the circulation of Enlightenment philosophy books. Given that large parts of the population were unable to read complex philosophical texts, Enlightenment ideas could nonetheless spread because books were often read aloud and discussed in public spaces, allowing their content to reach a much broader audience. As Darnton (1982) highlights, books were frequently read aloud in communal spaces, allowing even the illiterate to access new ideas. At the same time, one could argue that the decisive factor was not the broad diffusion of ideas, but rather that a small, intellectually and organizationally capable minority read these books and translated their values into collective action. In either case, a greater local availability of these works would have facilitated the broader diffusion of Enlightenment ideals. Enlightenment ideas could, in such a context of crisis, de-legitimize the regime, convincing that changes toward a democracy may

be desirable. In this sense, philosophy books serve the role of the migrant described above; it presents a new credible perspective of governance which inspires the demand for change. Thus, the circulation of philosophy books could serve as a catalyst for the spread of revolutionary ideas.

H2: A higher local presence of Enlightenment philosophy books will lead to greater local involvement in the Revolution.

Another crucial mechanism through which Enlightenment ideas could have influenced support for the Revolution is the direct role of intellectuals in promoting and diffusing these ideas. While many members of the population were unlikely to have access to complex philosophical texts, intellectuals may have acted as key conduits for the dissemination of these ideas. It is plausible that they translated abstract concepts into more accessible forms through public discourse and personal interactions.

As Léon et al. (1970) suggests, one of the decisive aspects of the transmission of Enlightenment ideas could have been interpersonal contact. While books were important, the diffusion of these ideas might have relied heavily on face-to-face interactions within social networks, where ideas were debated, discussed, and adapted. These direct exchanges could have played a significant role in spreading Enlightenment principles, even in areas where reading rates were low or access to books was limited.

Building on this framework, the paper hypothesizes that areas with a larger presence of intellectuals may have experienced greater support for revolutionary ideals. Intellectuals, by engaging with their social networks, could have played a central role in facilitating the diffusion of Enlightenment values.

H3: A higher local presence of intellectuals leads to greater local involvement in the Revolution.

4. DATA AND OPERATIONALIZATION

This paper uses various data sources that have already been exploited in the literature, as well as developing new ones. This section provides an overview of the main variables that will be used in the project.

4.1. Independent Variables

One of the main datasets that will be used in this project is the Famous People dataset (Laouenan et al., 2022). This data set is made up of information about the biographies of famous people, using data from their Wikipedia pages. From this data, this project will use information about their professions to derive a working category called intellectuals. This will be done following different methodologies. First, it will use the classification already present in the dataset (Culture core and Culture periphery). [1]

Regarding the geographical component, the dataset gives information about the coordinates place of birth and the place of death, which will be used to match individuals to the area of reference used in the analysis. I use the place of death as the main specification, as standard in the literature (Lecce et al., 2021; Serafinelli & Tabellini, 2022), but I also check the robustness of the results with the place of birth.

Regarding potential sources of bias in these data, while it is arguable that these sources contain mostly information about Western countries as well as other sources of bias concerning the over-representation of the current period versus more historical periods, this should not affect this paper. This is because this paper focuses only on one country and on just a single period, so the potential bias should be homogeneous among the units.

Concerning books, the main source of information is the French Book Trade in Enlightenment Europe project (Burrows, 2020). This project is a comprehensive initiative aimed at mapping the circulation of books and ideas during the Enlightenment, with a particular focus on the *Société Typographique de Neuchâtel* (STN). From this source, this project will collect data on key book categories for both hypotheses. For both hypotheses, I use the Parisian categorization contained on the website. For the first, I take novels, whereas for the second, I will take into consideration the philosophy category.

[1] The professions included in the intellectuals variable are the following: academic, academie, economist, économiste, essayist, historian, historiography, home de letters, man of letters, novelist, philosopher, philosopher, philosophie, professor, professeur, scholar, writer

4.2. Dependent Variables

Regarding the dependent variable, this project uses a series of variables used in the quantitative literature for measuring support for change and the revolution in general (Ottinger & Rosenberger, 2023; Squicciarini & Voigtländer, 2016; Waldinger, 2024).

To that end, I employ several complementary measures that reflect different forms of revolutionary engagement. First, I use data on the presence of early political societies, digitised from Boutry et al. (1992) and used by Ottinger and Rosenberger (2023), which serve as a proxy for organized political participation and engagement with revolutionary ideas. Second, I include anti-feudal revolts that occurred in 1789, to capture more spontaneous and grassroots expressions of revolutionary fervor (Chambru & Maneuvrier-Hervieu, 2024). Third, I draw on the share of refractory clergy in each area as a negative indicator of revolutionary alignment—areas with a high proportion of clergy refusing to swear loyalty to the Civil Constitution of the Clergy can be interpreted as less supportive of the Revolution (Squicciarini, 2020). Together, these indicators aim to capture a broad spectrum of revolutionary involvement, from organized activism to popular uprisings and resistance to the *ancien régime*.

In addition to these measures, I also draw on the Cahiers de Doléances, which have traditionally been used to capture local demands for reform (Squicciarini & Voigtländer, 2016; Waldinger, 2024). These documents may reflect early revolutionary sentiment, although the extent of participation by lower-level members of the Third Estate in their drafting is debated, as is their correspondence with preparatory cahiers (Hyslop, 1934). To extract systematic insights from them, I digitised data from Hyslop (1934) and coded the presence of reform demands as binary indicators. These dummies are disaggregated by the social order of the authoring group—clergy, nobility, and third estate—allowing for a more nuanced interpretation of how demands for change aligned with the proposed theoretical mechanisms.

4.3. Controls

The main potential avenue of endogeneity in the paper is economic prosperity. While some points against this relationship will be discussed later, this issue still needs to be addressed empirically. This will be done using a battery of usual controls in the economic history literature. The first controls are the population and the urban population [2], com-

[2] Urban population is compiled considering as cities places that have more than 5000 inhabitants (Bairoch et al., 1991; Ottinger & Rosenberger, 2023)

piled from the population census of 1793 (Degrave et al., 2023). Secondly, another control that is standardly used to look at issues of economic prosperity is the average height of soldiers [3] (Komlos, 2003).

Another issue that could be brought forward is the importance of literacy, as multiple variables used in the analysis are books. In particular, it could be a problem given Markoff (1986) findings on the different nature of violence. While its mechanisms still focus on the Enlightenment, the two works explore the same linkage; this potential source of endogeneity is easily controllable by using the data on literacy rates, which are publicly available (Squicciarini & Voigtländer, 2015).

Moreover, one concern may emerge from religiosity. It is acknowledged in the literature (Lecce et al., 2021) that the birth of scientists is negatively associated with the level of religiosity in the area; if a similar mechanism were true in general for intellectuals, it could cause endogeneity both by decreasing the presence of intellectuals and the support for the revolution. To control this issue, the paper will use one of the controls presence of bishopric cities (Lecce et al., 2021).

Other controls that will be used in the analysis are ruggedness (Nunn & Puga, 2012), and the presence of Roman roads (McCormick et al., 2013). Finally, I also coded family types at the department level following Todd (1990) maps.

[3] From this source, I take the geolocation of their place of birth from which I compile the average at the district level

5. HISTORICAL SECTION

This section provides a brief historical overview of the origins of the data used in the project, placing it within its broader historical context.

5.1. Societé typographique de Neuchâtel

Created in 1769, the Societé typographique de Neuchâtel (STN) quickly grew as one of the biggest printing facilities in Europe. Selling over the course of the 25 years of its existence over 400000 books, about a fifth of the book production in Switzerland in the period (Buringh & Van Zanden, 2009). Thanks to the restrictive norms of the French book market (Inderwildi, 2008), the STN rapidly became one of the largest sellers of the Enlightenment in France during the period, with around 150000 (Curran, 2018). This position was facilitated by its geographical position close to the French border, but enjoying the relatively freer set of rules of the Swiss market. While selling books of different categories, the STN tended to specialise in the sell of unique editions, thanks to its unique set of relations with authors of the Enlightenment preferring to print their books not in France, among whom Voltaire, as well as books coming from Switzerland over which it had an advantage compared to the other editors because of the size of its printing facilities, 12 printing press(Curran, 2018). Comparing this with the standard of the French editors, the STN would classify among one of the biggest printing facilities, as accounts say that only 3 editing facilities had 5 or more printing presses.

After reaching a peak of sales to the French market in 1782 with over 70000 books sold only in the year, the activity of the STN fell into crisis. This was probably due to a royal decree in 1783 which stated that every book coming from abroad had to first go to Paris to be controlled before going to the final destination (Curran, 2018). This tightening of the French rules significantly damaged the STN, which lost its main trading country. Although it attempted to open new trade routes to the north of Europe, it was never able to succeed, eventually closing in 1793.

5.2. Political societies

Initially known as clubs, political societies represent a unique form of grassroots mobilization during the French Revolution. The first one was founded in Paris in 1789. First, just tolerated and then being granted a more favorable legal framework in 1791 by the *As-*

semblée Générale, political societies were the way through which citizens discussed and remained up to date with the development of the revolution. Initially, this phenomenon was prevalent in urban areas, but it diffused with time to even smaller geographical entities (Boutry et al., 1992). The rise of Napoleon Bonaparte marked the end of this experiment in popular political participation. Viewing such societies as sources of factionalism and potential unrest, the Napoleonic regime moved swiftly to suppress independent political organization. By the early 1800s, political societies had been effectively dismantled, and a new order replaced the deliberative, decentralized model of revolutionary politics.

5.3. Refractory Clergy

Before the advent of the French Revolution, the Catholic church in France had a favourable legal status, being represented in the national assembly and having the power to collect taxes (Bonin & Langlois, 1987). One of the changes brought about by the French Revolution was the attempt to reduce the influence of the Church by making it an element at the service of the state (Squicciarini, 2020; Woell, 2015). One of the acts toward this goal was the civil constitution of the church by the National Assembly, demanding that clergymen take an oath of loyalty toward the French State. This act generated a schism within the church between a part of the clergy becoming subject to the French State and a part refusing, the refractory clergy. This schism divided the French population, generating a large backlash in the most religious parts of the population (Woell, 2015).

5.4. Cahiers de Doleances

The Cahiers de Doleances, lists of grievances, represent a unique source of information on modern history. Corresponding to an old tradition of the French state, when the French king called the *États Généraux*, members of the three estates would compile these lists of grievances at the local level, preparatory cahiers, which would then be compiled into a unique list for each estate at the electoral level, the *bailliage* (Hyslop, 1934). After the elections of the representative to the Assembly, the delegates would bring the *Cahiers* to the assembly to make the complaints of the population heard by the King.

Generally inconclusive because of its parochial demands, the 1789 *Cahiers* represent an exception because of the presence of multiple issues of national interest, notwithstanding the prevalence of demand for the reduction of taxes (Shapiro et al., 1998). These documents played a critical role in shaping the discursive framework of the Revolution, offering a rare snapshot of political consciousness across social classes on the eve of

radical change. The *Cahiers* not only catalogued grievances but also revealed a growing political imagination that helped lay the groundwork for revolutionary demands for rights, representation, and reform.

6. METHODOLOGY

6.1. Empirical Strategy

This paper aims to grasp causality using modern causal inference techniques; however, it acknowledges the difficulties and limitations that are behind this project, at least at this stage. For this reason, the baseline model will be a simple multivariate regression analysis, including the controls put forward in the previous sections. This first step represents the first baseline estimation. To provide further evidence of the relationship, this project also conducts placebo tests. One example of this is to use the outcomes of the *Cahiers de Doleances* that should not be, in theory, related to predicaments of the Enlightenment, such as economic liberalism, as already done in the literature (Squicciarini & Voigtländer, 2016). If these placebo tests show null results, they will make the entire analysis more credible.

Using Sacred texts, I conduct another placebo to see whether the effects still appear. A null result would, in this case, ensure that any other unobservable related to book reading is not confounding the analysis.

Moving beyond the regression analysis, this project also instruments the presence of books by using distance to Neuchâtel. The idea for this instrument comes from the gravitational model of trade (Bergstrand, 1985; Tinbergen, 1962) commonly used in trade theory. According to the model, distance should predict the amount of trade between two elements. If this is the case, distance should strongly predict the presence of books from the STN. At the same time, more distance may also predict a higher risk of the book getting confiscated, as most of the books traded by the STN were illegal. This should provide a random source of variation to exploit for the analysis. To clarify the logic of the instrument in an experimental context, let's consider a scenario involving "always-takers." In this case, the paper aims to assess the impact of reading a book or having indirect contact with an intellectual. The challenge lies in the fact that both reading a book and conversing with someone could be non-random behaviors. Some individuals may already hold certain values, regardless of any external influences. By using an instrument, however, the study can randomize the likelihood of purchasing a book from the STN, thus providing a way to isolate the causal effect.

6.2. Discussion of potential issues of Endogeneity and Selection Bias

An obvious potential critique of the mechanisms highlighted in the paper is that this process is endogenous to value change caused by modernization (Chartier, 1991; Inglehart & Welzel, 2005; Lipset, 1959). More precisely, one might argue that the request for change stems from the emergence of a new bourgeoisie demanding reforms. This may be linked to the presence of intellectuals, as there is some evidence that the economic development of an area may lead to a greater presence of intellectuals (De La Croix et al., 2023; Serafinelli & Tabellini, 2022). However, there are historical reasons to believe that this is not the case. First of all, the lack of models from which a growing bourgeois class could have taken inspiration to achieve the results of the French Revolution requires a new and unique formulation of the model to be pursued, which, according to Israel (2015), stems from the work of the Radical Enlightenment. Secondly, the association between this intellectual elite and a richer area is dubious in the case of France. Previous research (Squicciarini & Voigtländer, 2015) demonstrated the lack of a relationship between the presence of intellectuals and economic growth before the Revolution in France. This adds doubt to the utility of the modernization hypothesis (Lipset, 1959) to explain support for the revolution. Finally, given the availability of various proxies of economic growth, the question of endogeneity becomes an empirical one, as it is possible to control for this potential pathway.

Parallel to this issue, there is the potential for selection bias. This would happen if the mechanisms causing the migration of intellectuals from one area to another could simultaneously predict support for the revolution. One selection mechanism would be religiosity. Religious areas may be more conservative, hence supporting the revolution less, and at the same time causing a higher emigration of intellectuals. This second part has been empirically tested by Lecce et al. (2021). The authors discard the possibility of such effects. Concerning other potential pull and push factors, De La Croix et al. (2023) and Serafinelli and Tabellini (2022) mention the role of money and the presence of prestigious universities. All of these are potentially accountable empirically.

7. DESCRIPTIVE DATA

This section presents the data and describes the variables thoroughly, which will be used later in regression analysis. Figures 7.1 and 7.2 describe the spatial distribution of the books delivered by the STN across several categories. The data is presented using books per capita as the main variable. A visual inspection suggests that the instrument is working as expected: cities closer to the Swiss border, where the STN was headquartered, received a greater number of books. While there is some variation, this trend is consistent across the two categories. That said, the distribution also reveals a few notable outliers—such as Loudun—which warrant further exploration. Looking deeper at this observation, it seems like this is driven by the presence of a very important single buyer, *Malherbe, l'aîné*, whose role is very important for the late activity of the STN after the decree of 1783 aiming to destroy the market of illegal books in France (Burrows, 2015). As a single client, he bought between 7999 and 8215 books, which would later be resold.

Figure 7.1

Spatial diffusion of novels from the STN

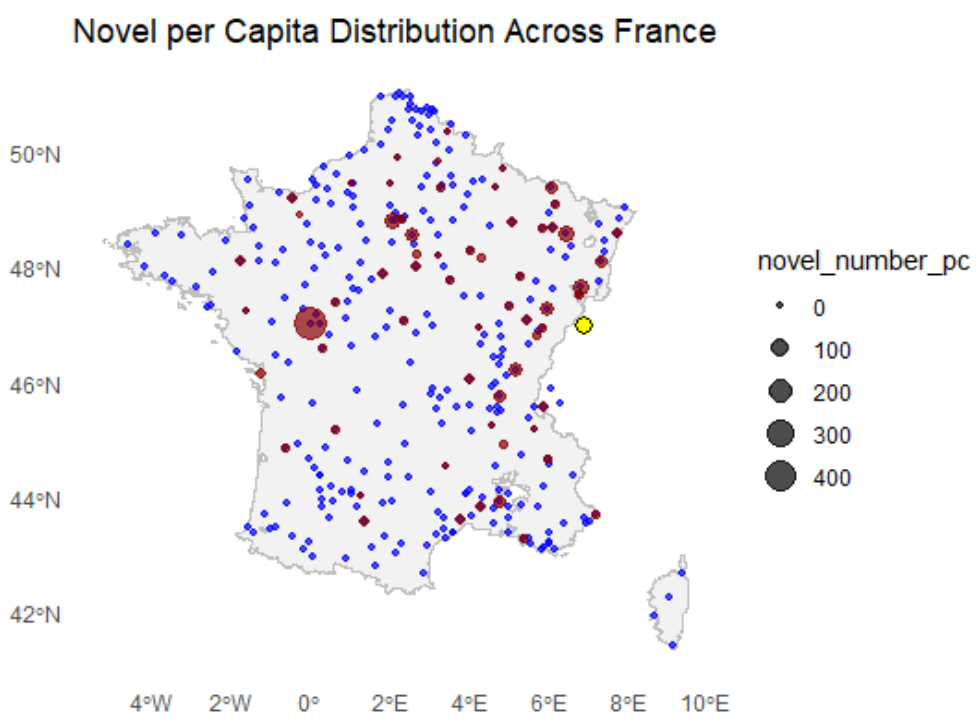
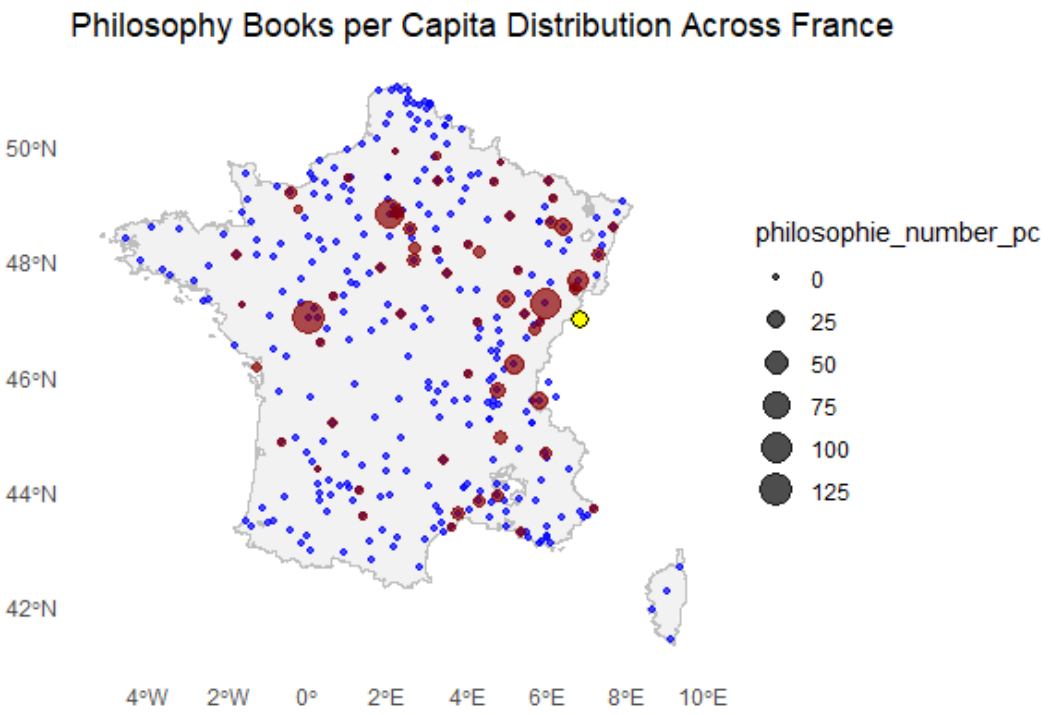


Figure 7.2

Spatial diffusion of philosophy books from the STN



Next, I turn to the geographical distribution of political societies, including Jacobin and non-Jacobin organizations. The dataset contains 6,530 societies established between 1789 and 1794, with 948 identified as Jacobin. These were mapped at the municipality level, showing that 5,186 municipalities hosted at least one such society. Relative to France's current 34,965 municipalities, this suggests that about one in seven towns had a political society during the Revolution. Figure 3 shows the time trends: Jacobin societies initially formed the majority, but their growth tapered off after 1792 with Robespierre's rise. In contrast, non-Jacobin societies expanded rapidly, especially by 1794. Figures 7.4 and 7.5 illustrate distinct regional clustering, particularly in the Paris area and the South-East.

Parallel to this, I also report the main descriptive statistics for all key variables used in the analysis are presented in Table 9.1 in the Appendix.

Figure 7.3

Time trends for Jacobin and non Jacobin societies

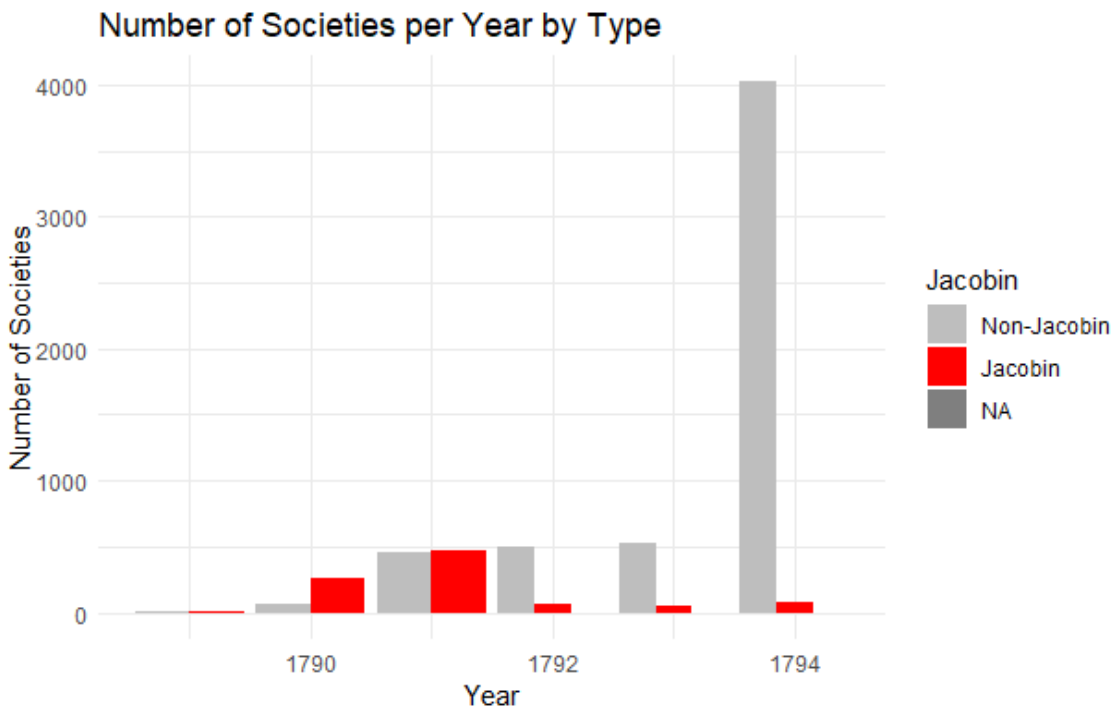


Figure 7.4

Geographical distribution of Jacobin societies

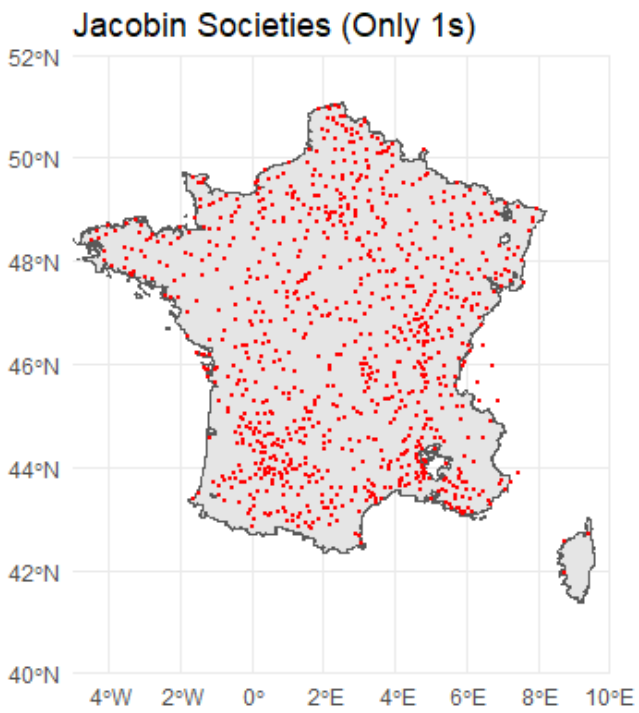
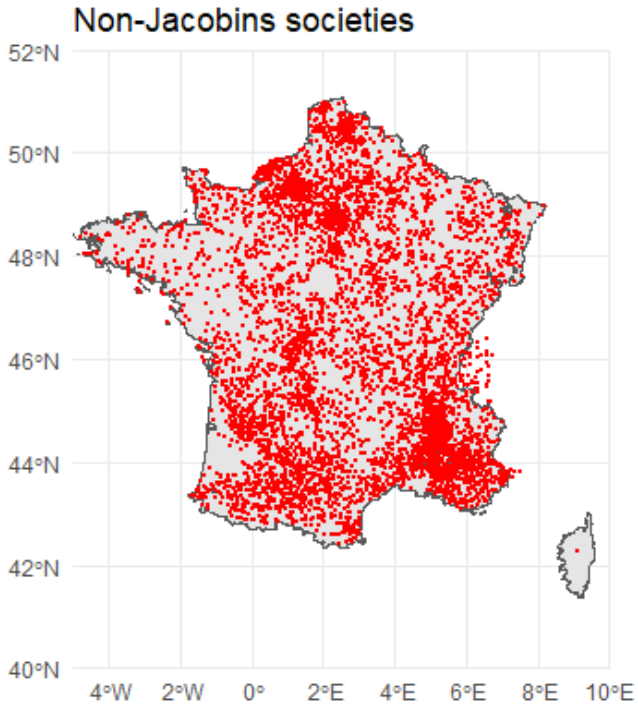


Figure 7.5

Geographical distribution of non-Jacobins societies



8. RESULTS

8.1. Main Specifications

This section presents the empirical results using districts as the unit of observation. Ordinary Least Squares models (OLS) and generalised linear models (GLM) are estimated with standard control variables and department-level fixed effects in order to mitigate potential concerns related to endogeneity and selection bias. Each hypothesis is tested using multiple dependent variables: a binary indicator for the presence of district-level conflicts, the share of refractory clergy as a proxy for clerical resistance, and the number of non-Jacobin political societies, which is selected for its greater within-sample variation [4]. Given the count structure of the third dependent variable, the specification for political societies is estimated using a Negative Binomial model rather than a standard OLS regression.

The results, presented in Table 8.1, reveal a stable and theoretically consistent relationship between the presence of intellectuals and revolutionary outcomes. Specifically, intellectual presence is positively and significantly associated with both the incidence of conflict and the formation of political societies. However, no statistically significant relationship emerges in models explaining clerical resistance. This null finding may reflect the clergy's institutional autonomy from local intellectual environments or the impact of broader religious and hierarchical dynamics not accounted for in the model.

Regarding the control variables, the results also yield insightful implications. The negative effect of literacy, while initially counterintuitive, aligns with existing scholarship. As shown in prior work, such as Squicciarini and Voigtländer (2016), literacy alone does not positively predict radical political behavior. This finding also supports the interpretation that literacy functioned primarily as a proxy for Enlightenment exposure. Once Enlightenment diffusion is directly measured, literacy no longer exerts an independent effect. This aligns with the logic of earlier work, such as Markoff (1986), who argued that the observed relationship between literacy and anti-feudal mobilization was likely driven by ideational channels rather than literacy itself. The economic controls included to test modernization-based explanations are largely insignificant, except for the total population, and, in some

[4] Using Jacobin societies can also be historically problematic: In fact, the radicalization of the Jacobins movement begins with the attempt of the King to flee France on the 19th June of 1791. To have a sense of the impact of the Enlightenment on the Jacobins' movement before the radicalisation, one would have to select only societies born before 1791, which would limit even more the sample.

cases, even negatively associated with revolutionary outcomes. These findings lend support to recent historiographical arguments challenging traditional bourgeoisie-centered interpretations of revolutionary mobilization, such as those advanced by Lucas (1973). In addition, the negative relationship between urbanization and the number of non-Jacobin societies appears to be a compositional artifact: highly urbanized districts consist of fewer distinct territorial units, which mechanically limits the number of possible societies that can be formed.

In the baseline OLS specifications, the variable capturing the presence of books is consistently not in line with the hypothesis across all models. Given that the analysis aims to estimate the direct effect of book availability on revolutionary outcomes, this null finding raises concerns about potential attenuation due to measurement error or endogeneity. To address this, the next section introduces an instrumental variable strategy using distance from Neuchâtel.

8.2. Exploring the Mechanism with the *Cahiers de Doleances*

To further explore the mechanism through which intellectual presence influenced revolutionary outcomes, I examine the content of the Cahiers de Doléances as a proxy for the nature of local political demands on the eve of the Revolution. In this analysis, I test whether the local presence of intellectuals is associated with demands for democratisation versus economic demands. The results, presented in Table 8.2, indicate a strong and positive association between intellectual presence and the prevalence of democratic demands, while the relationship with economic grievances remains weak and statistically insignificant.

This pattern suggests that intellectuals played a key role in shaping the ideological and political content of revolutionary mobilization, rather than diffusing economic values or directly promoting material reform agendas. Importantly, since political and economic reforms were often bundled during the Revolution, this finding provides evidence against the interpretation that political demands were merely derivative of underlying economic transformations. Instead, it supports a sequential view in which the articulation of political claims — potentially inspired by Enlightenment discourse — preceded and perhaps enabled the development of modern economic institutions. In this context, intellectual activity appears to have contributed primarily to the politicization of public life, helping to construct the frameworks through which broader social and institutional change became thinkable.

	Dependent variable:								
	Conflict Dummy			Share Refractory			Non-Jacobin Societies		
	OLS			*OLS*			*Negative Binomial*		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log(Philosopher Deaths + 1)	0.163***			−0.007			0.150**		
	(0.046)			(0.022)			(0.069)		
Log(Novels Published + 1)		0.009			0.013*			0.027	
		(0.015)			(0.007)			(0.022)	
Log(Philosophy Books + 1)			0.013			0.015**			0.033
			(0.016)			(0.007)			(0.024)
Log(Total Population)	0.122*	0.159**	0.157**	0.057*	0.052	0.051	0.666***	0.703***	0.699***
	(0.070)	(0.070)	(0.070)	(0.033)	(0.033)	(0.033)	(0.108)	(0.107)	(0.107)
Male Literacy Rate (1786)	0.324	0.183	0.186	0.299*	0.313**	0.315**	−1.592***	−1.663***	−1.656***
	(0.333)	(0.338)	(0.338)	(0.160)	(0.157)	(0.157)	(0.470)	(0.469)	(0.468)
Mean Ruggedness	0.004	0.004	0.004	0.001	0.002	0.002	0.007	0.009	0.009
	(0.011)	(0.011)	(0.011)	(0.005)	(0.005)	(0.005)	(0.015)	(0.015)	(0.015)
Mean Height	−0.007	−0.007	−0.007	−0.004	−0.004	−0.004	0.020	0.023	0.022
	(0.013)	(0.013)	(0.013)	(0.006)	(0.006)	(0.006)	(0.021)	(0.021)	(0.021)
Urbanization Rate	−0.133	−0.001	−0.009	0.028	−0.039	−0.038	−0.537^{+}	−0.505^{+}	−0.514^{+}
	(0.184)	(0.193)	(0.191)	(0.088)	(0.090)	(0.089)	(0.283)	(0.293)	(0.289)
Egalitarian × Nuclear Family	−0.028	−0.014	−0.014	−0.099	−0.099	−0.099	−0.034	−0.014	−0.014
	(0.186)	(0.190)	(0.190)	(0.089)	(0.089)	(0.089)	(0.274)	(0.276)	(0.276)
Bisphoric Dummy	0.029	0.034	0.034	0.020	0.021	0.020	−0.116	−0.117	−0.118
	(0.047)	(0.048)	(0.048)	(0.023)	(0.022)	(0.022)	(0.072)	(0.072)	(0.072)
Roman Road Present	−0.060	−0.063	−0.064	−0.013	−0.015	−0.015	0.145**	0.138*	0.137*
	(0.044)	(0.045)	(0.045)	(0.021)	(0.021)	(0.021)	(0.070)	(0.071)	(0.071)
Constant	−0.063	−0.403	−0.396	0.361	0.282	0.307	−7.745*	−8.502**	−8.441**
	(2.437)	(2.490)	(2.488)	(1.167)	(1.160)	(1.159)	(4.013)	(4.028)	(4.021)
Department FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	353	353	353	353	353	353	353	353	353
R^2	0.544	0.525	0.525	0.763	0.766	0.766			
Adjusted R^2	0.429	0.405	0.405	0.703	0.707	0.707			
Log Likelihood							−992.138	−993.803	−993.558
θ							9.111*** (1.354)	8.999*** (1.336)	9.038*** (1.344)
AIC							2,128.275	2,131.607	2,131.115
Residual Std. Error (df = 281)	0.305	0.311	0.311	0.146	0.145	0.145			
F Statistic (df = 71; 281)	4.730***	4.371***	4.378***	12.718***	12.940***	12.961***			

Note: $^{+}$p<0.1; *p<0.05; **p<0.01; ***p<0.001

Table 8.1

Testing the three hypotheses with an OLS

	Dependent variable:	
	‘Cahiers that were most strongly Democratic Third Etat‘	‘Cahiers giving Both Mercantilist and Liberal Economic Demands Third Etat‘
	(1)	(2)
log(Intellectuals + 1)	0.184**	0.016
	(0.082)	(0.085)
log(Total Population)	0.073	0.010
	(0.086)	(0.090)
Urbanization Rate	−0.198	0.448
	(0.356)	(0.371)
Bisphoric Dummy	−0.004	0.037
	(0.103)	(0.108)
Mean Height	−0.015	0.026
	(0.046)	(0.048)
Constant	2.400	−3.667
	(7.870)	(8.196)
Observations	109	109
R^2	0.525	0.476
Pays d’Election FE	Yes	Yes
Note:		*p<0.1; **p<0.05; ***p<0.01

Table 8.2

Intellectual Presence and the Nature of Demands in the Cahiers de Doléances (Third Estate)

8.3. Instrumental Variable Approach

To estimate the causal impact of book presence on revolutionary outcomes, an instrumental variable (IV) strategy is employed. The baseline OLS models consistently find the book presence variable to be insignificant, which may be due to measurement error, omitted variables, or inappropriate model specification. Because book presence is the main variable of interest, this section aims to address these concerns using an IV approach.

The instrument is based on a gravity model reflecting the spatial diffusion of books. Distance from Neuchâtel is used as an instrument for book presence in each district. The assumption is that districts farther from Neuchâtel have less access to books simply due to transportation and distribution constraints, providing a source of exogenous variation. The peripheral position of Neufchâtel,a small Swiss city of 5000 inhabitants, in French history ensures the validity of the exclusion restriction clause.

The strength of this instrument is assessed in the first-stage regressions, reported in Table 8.3. The F-statistics for the instrument exceed the conventional threshold of 10, with values of 13.77 when predicting novels and 15.05 when predicting philosophical works. These values indicate strong instruments and reduce concerns about weak instrument bias.

Coefficients for distance are negative and significant as theoretically expected.

In the second stage, the predicted values of book presence from the first stage are used to estimate its effect on all three dependent variables: district-level conflicts, clerical resistance, and the number of non-Jacobin political societies. The IV results, shown in Table 8.4, indicate statistically significant effects of book presence for all of the models except for the political societies, consistent with the theoretical expectations.

Importantly, the IV estimates for district-level conflicts and clerical resistance yield statistically significant effects, indicating that books played a meaningful role in shaping the response of the counter-revolutionary movements. By contrast, the re-estimated model for political societies—specified as a Negative Binomial regression to account for the count structure—fails to reach significance, suggesting a more limited or indirect effect of book circulation on the formation of non-Jacobin political organizations.

The divergence between the OLS and IV results warrants further consideration. One possibility is that the baseline OLS models fail to account for local religiosity, which may simultaneously limit the diffusion of Enlightenment books and reduce support for revolutionary change. In more devout districts, books promoting Enlightenment ideas may have been less available or actively resisted, while at the same time, religious conservatism could dampen revolutionary mobilization. This dual influence would bias OLS estimates downward by obscuring the true effect of book exposure. A second possibility relates to the nature of book readership in this period: Enlightenment texts were culturally fashionable, and their consumption was not necessarily limited to neutral individuals. If individuals who were ideologically resistant to the Enlightenment still read these works—out of curiosity, prestige, or social pressure—this kind of self-selection would attenuate the observed correlation in OLS. By using distance-based instruments, the IV approach arguably recovers the influence of exposure net of this selection, thus revealing a stronger underlying effect.

	Dependent variable:	
	log(philosophy_number + 1)	log(novels_number + 1)
	(1)	(2)
log(distance from Neufchâtel)	-0.491^{***}	-0.514^{***}
	(0.127)	(0.139)
log(Total population)	0.825^{***}	0.878^{***}
	(0.148)	(0.162)
Constant	-5.440^{***}	-5.826^{***}
	(1.573)	(1.724)
Observations	414	414
R^2	0.084	0.079
Note:		*p<0.1; **p<0.05; ***p<0.01

Table 8.3

First stages from the IV regression

	Dependent variable:					
	dummy_conflict		share_refractory		societies_Non_Jacobins	
	instrumental variable		*instrumental variable*		*IV with negative binomial at the second stage*	
	(1)	(2)	(3)	(4)	(5)	(6)
log(Philosophy books + 1)	0.555***		−0.158**			
	(0.160)		(0.066)			
log(Novels + 1)		0.530***		−0.151**		
		(0.159)		(0.064)		
log(predicted philosophy books + 1)					0.027	
					(0.047)	
log(predicted Novel + 1)						0.108
						(0.148)
log(Total Population)	−0.289**	−0.296**	0.270***	0.272***	0.348***	0.313**
	(0.141)	(0.148)	(0.058)	(0.060)	(0.115)	(0.138)
Constant	3.057**	3.123**	−2.392***	−2.411***	−1.411	−1.328
	(1.451)	(1.526)	(0.601)	(0.613)	(1.160)	(1.134)
Observations	414	414	414	414	414	414
Note:						*p<0.1; **p<0.05; ***p<0.01

Table 8.4

Second stage of the IV estimate

8.4. Robustness Checks

This section presents several robustness checks that strengthen the validity of the findings.

First, I re-estimate the instrumental variable (IV) models using a Poisson regression in the first stage, instead of ordinary least squares (OLS). This adjustment follows standard practice in trade theory, where Poisson pseudo-maximum likelihood is often preferred for estimating gravity models due to its robustness to heteroscedasticity and its ability to avoid bias that OLS might introduce when applied to log-linearized models (Silva & Tenreyro, 2006). The results, displayed in the Appendix in table 9.2, are consistent with those from the original specification.

Second, I re-coded the presence of intellectuals using place of birth instead of place of death. This adjustment, displayed in table 9.3, renders the results insignificant in the

first two specifications, while remaining consistent for political societies. Crucially, this difference underscores the importance of migration. Unlike place of birth, which may be relatively random, intellectual clusters often form through movement, influencing revolutionary outcomes. Migration was especially prevalent among prominent intellectuals, potentially amplifying their impact (De La Croix et al., 2023). Given these considerations, the place of death specification is preferable as it better captures the spatial concentration and influence of intellectuals during the revolutionary period.

Third, I conduct a placebo test using the diffusion of sacred texts—rather than Enlightenment literature—as the main explanatory variable. Given the lack of a strong first-stage relationship in the instrumental variable model, I focus on OLS specifications. Results in table 9.4 show that sacred texts are positively associated with clerical refractoriness and in the opposite direction of what is expected from the hypothesis, while their relationship with conflict and the formation of political societies is statistically insignificant. For further comparison, I also present the OLS results for political societies using sacred texts, which for intellectuals are marginally significant at the 10% level. The comparatively smaller coefficient in this case illustrates the substantive difference between the impact of novels and philosophy books versus religious material.

As an additional test, I replace the number of intellectuals with a different category— famous people from the Military— to examine whether the observed effects are attributable to intellectual influence per se, or rather reflect general social prominence or fame. Results in table 9.5 are null in all of the specifications and stable to taking birth and death years. This evidence rejects the idea that the effects are in reality related to underlying trends in fame.

9. CONCLUSION

This paper has sought to empirically assess the long-standing claim that the French Revolution was fundamentally shaped by intellectual currents. By leveraging sub-national variation in revolutionary support and local exposure to Enlightenment ideas, it provides evidence in favor of the Tocquevillian thesis: that ideas, not merely material grievances, played a central role in mobilizing revolutionary change. The analysis suggests that districts more deeply exposed to Enlightenment thought were more likely to experience conflict and to host non-Jacobin political societies, reinforcing the notion that ideational diffusion shaped patterns of political engagement.

Beyond its specific empirical findings, this paper underscores the value of bridging intellectual history with quantitative social science. By treating Enlightenment thought not only as a cultural artifact but also as a measurable influence, the study opens space for new empirical strategies in assessing the impact of ideas.

While these results remain preliminary, and further work is required to clarify causal mechanisms and rule out alternative explanations, the evidence points toward the enduring relevance of ideas in explaining one of the most transformative episodes of modern history. Econometrically, concerns towards the general validity of the results for books are ensured by the addition of fixed effects, which, by making the comparison only between entities within the same districts, alleviates concerns. This specification also tends to be more punishing against the hypothesis as it significantly diminishes the amount of variation that can be used in the analysis.

On a more general level, this study seeks to underscore the significance of high ideas in the analysis of major political transformations, challenging the view that ideas are merely epiphenomenal or fully endogenous to material conditions. Instead, it argues that ideas possess a degree of autonomy and causal power in shaping historical outcomes. More specifically, this paper presents the French Revolution as a compelling case study, insofar as it anticipates social and political transformations that are more commonly attributed to the processes of industrialization. By occurring prior to widespread industrial development, the Revolution offers a unique context in which to isolate the role of ideas, rather than economic modernization, as a primary catalyst of systemic change. Coherent with the general narrative, even indicators of proto-economic development tend to be uncorrelated with the support for the Revolution.

Concerning the limitations of this study, the most significant constraint arises from

data availability. Given the historical nature of the inquiry, obtaining reliable and comprehensive data at a fine-grained spatial level, such as the district level, is inherently difficult. Moreover, while the analysis incorporates a range of controls and robustness checks, the absence of a fully exogenous identification strategy limits the strength of causal claims, particularly in evaluating the impact of intellectual presence. Without a source of random variation or a natural experiment, it remains challenging to fully disentangle ideational influence from confounding local characteristics, as well as fully capturing the mechanism.

Finally, the study remains intentionally agnostic regarding the precise mechanisms of ideational transmission. Although the interpretation that books and intellectuals fostered revolutionary sentiment through the diffusion of Enlightenment ideas is consistent with established historical narratives, alternative mechanisms could plausibly account for the observed patterns. For instance, as Fishman (2004) suggests, intellectual exposure may have furnished individuals with new discursive repertoires and rhetorical tools, which in turn allowed them to articulate and channel existing grievances more effectively, thus facilitating broader mobilization. Testing such mechanisms, however, would require richer data on individual cognition and communication practices, which are beyond the scope of the current dataset.

Notwithstanding these limitations, this paper, at a minimum, shows a consistent correlation between ideas and greater support for the Revolution, contributing to the specific debate on the intellectual origins of the French Revolution.

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APPENDIX

Variable	Mean	SD	Min	Max	N
Number of novels	30.572	171.140	0	2,335	416
Number of philosophy books	25.038	208.805	0	3,473	416
Number of philosophers (dead)	0.489	1.031	0	7	417
Number of sacred books	10.635	85.164	0	1,451	417
Share of refractory clergy	0.431	0.267	0	1	417
Number of non-Jacobin societies	10.942	9.929	0	78	417
Number of Jacobin societies	1.823	1.571	0	9	417
Conflict (dummy)	0.213	0.410	0	1	417
Total population	49,215.310	25,382.030	14,320	232,528	415
Urbanization rate	0.097	0.141	0	0.933	415
Mean height (cm)	169.249	1.640	162.400	177.290	410
Male literacy (1786)	0.373	0.244	0	0.900	370
Mean ruggedness	3.503	3.373	0.480	21.735	416
Bisphoric (dummy)	0.273	0.446	0	1	417
Egalitarian family (dummy)	0.424	0.495	0	1	403
Nuclear family (dummy)	0.521	0.500	0	1	403
Has Roman road (dummy)	0.685	0.465	0	1	416

Table 9.1

Descriptive Statistics

	Dependent variable:			
	Conflict Measures		Share of Refractory Clergy	
	(1) Conflict (Philosophy)	(2) Conflict (Novels)	(3) Resistance (Philosophy)	(4) Resistance (Novels)
log(Predicted Philosophy Books +1)	0.169**		−0.042**	
	(0.023)		(0.015)	
log(Predicted Novels +1)		0.527**		−0.148**
		(0.071)		(0.047)
log(Total Population)	−0.183**	−0.294**	0.231**	0.270**
	(0.055)	(0.066)	(0.037)	(0.044)
Constant	1.752**	1.584**	−1.934**	−1.961**
	(0.557)	(0.546)	(0.369)	(0.360)
Observations	414	414	414	414
R^2	0.127	0.126	0.094	0.098
Note:				+p<0.1; *p<0.05; **p<0.01

Table 9.2

Second Stage IV Estimates, replicating the first stage with a Poisson

	Dependent variable:		
	Conflict Dummy	Share Refractory	Non-Jacobin Societies
	OLS	*OLS*	*Negative Binomial*
	(1)	(2)	(3)
log(Intellectuals at Birth +1)	0.008	0.021	0.164**
	(0.039)	(0.018)	(0.060)
log(Total Population)	0.157*	0.046	0.636**
	(0.072)	(0.034)	(0.110)
Male Literacy Rate (1786)	0.191	0.342*	−1.435**
	(0.345)	(0.161)	(0.478)
Ruggedness	0.003	0.001	0.009
	(0.011)	(0.005)	(0.015)
Mean Height	−0.007	−0.004	0.020
	(0.013)	(0.006)	(0.021)
Urbanization Rate	0.025	−0.021	−0.692*
	(0.197)	(0.092)	(0.295)
Egalitarian × Nuclear Family	−0.014	−0.099	−0.017
	(0.190)	(0.089)	(0.275)
Biphoric Region Dummy	0.034	0.020	−0.111
	(0.048)	(0.022)	(0.072)
Roman Road Presence	−0.061	−0.012	0.145*
	(0.045)	(0.021)	(0.070)
Constant	−0.313	0.447	−7.391+
	(2.493)	(1.166)	(4.029)
Department Fixed Effects	Yes	Yes	Yes
Observations	353	353	353
R^2	0.524	0.764	
Note:			+p<0.1; *p<0.05; **p<0.01

Table 9.3

Birth Replication

	Dependent variable:		
	Conflict Dummy	Share Refractory	Non-Jacobin Societies
	OLS	*OLS*	*Negative Binomial*
	(1)	(2)	(3)
log(Sacred Books +1)	−0.002	0.016*	−0.00002
	(0.020)	(0.009)	(0.029)
log(total Population)	0.161*	0.052	0.709**
	(0.071)	(0.033)	(0.108)
Male Literacy Rate (1786)	0.177	0.307*	−1.695**
	(0.338)	(0.158)	(0.471)
Ruggedness	0.003	0.002	0.007
	(0.011)	(0.005)	(0.015)
Mean Height	−0.007	−0.004	0.021
	(0.013)	(0.006)	(0.022)
Urbanization Rate	0.045	−0.016	−0.363
	(0.188)	(0.088)	(0.283)
Egalitarian × Nuclear Family	−0.014	−0.100	−0.017
	(0.190)	(0.089)	(0.277)
Biphoric Dummy	0.034	0.017	−0.118
	(0.048)	(0.022)	(0.072)
Roman Road Presence	−0.062	−0.012	0.143*
	(0.045)	(0.021)	(0.071)
Constant	−0.334	0.316	−8.346*
	(2.491)	(1.162)	(4.047)
Observations	353	353	353
R^2	0.524	0.765	

Note: $^{+}p<0.1$; $^{*}p<0.05$; $^{**}p<0.01$

Table 9.4

Changing Books to Sacred Books

	Dependent variable:		
	Conflict Dummy	Share Refractory	Non-Jacobin Societies
	(1)	(2)	(3)
log(Leadership Deaths +1)	0.002	−0.001	0.552
	(0.033)	(0.016)	(0.704)
log(Total Population)	0.160*	0.055	8.042**
	(0.072)	(0.034)	(1.525)
Male Literacy Rate (1786)	0.176	0.305*	−22.989**
	(0.339)	(0.159)	(7.142)
Ruggedness	0.003	0.001	0.067
	(0.011)	(0.005)	(0.227)
Mean Height	−0.007	−0.004	−0.022
	(0.013)	(0.006)	(0.280)
Urbanization Rate	0.037	0.022	−7.753+
	(0.201)	(0.094)	(4.230)
Egalitarian × Nuclear Family	−0.014	−0.100	−1.905
	(0.190)	(0.089)	(4.008)
Biphoric Dummy	0.033	0.020	−1.614
	(0.048)	(0.023)	(1.013)
Roman Road Presence	−0.062	−0.012	1.152
	(0.045)	(0.021)	(0.940)
Constant	−0.329	0.369	−63.938
	(2.501)	(1.172)	(52.628)
Observations	353	353	353
R^2	0.524	0.763	0.669

Note: +p<0.1; *p<0.05; **p<0.01

Table 9.5

Changing the Category from Intellectuals to Leadership

	Dependent variable: Non-Jacobin Societies		
	(1)	(2)	(3)
	OLS	*OLS*	*OLS*
log(Novels + 1))	0.574*		
	(0.310)		
log(Philosophy Books + 1)		0.644*	
		(0.336)	
log(Sacred Books + 1)			0.018
			(0.424)
log(Total Population)	8.208***	8.146***	8.319***
	(1.476)	(1.477)	(1.487)
Literacy Males (1786)	−22.139***	−22.060***	−22.502***
	(7.082)	(7.080)	(7.123)
Mean Ruggedness	0.119	0.117	0.073
	(0.228)	(0.227)	(0.228)
Mean Height	0.011	0.009	−0.014
	(0.279)	(0.279)	(0.281)
Urbanization Rate	−8.878**	−8.808**	−6.374
	(4.042)	(4.012)	(3.955)
Egalitarian Family $\times NuclearFamily$	−1.889	−1.883	−1.924
	(3.988)	(3.986)	(4.012)
Bisphoric Dummy	−1.535	−1.562	−1.568
	(1.006)	(1.005)	(1.015)
Roman Road	1.024	1.013	1.125
	(0.936)	(0.936)	(0.940)
Constant	−71.649	−70.577	−67.889
	(52.174)	(52.130)	(52.473)
Observations	353	353	353
R^2	0.672	0.673	0.668

Note: $^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$

Table 9.6

Sacred texts vs Novels and Philosophy Books OLS